Boundary-Free Endpoint Unboundedness on Simply Connected Planar Domains

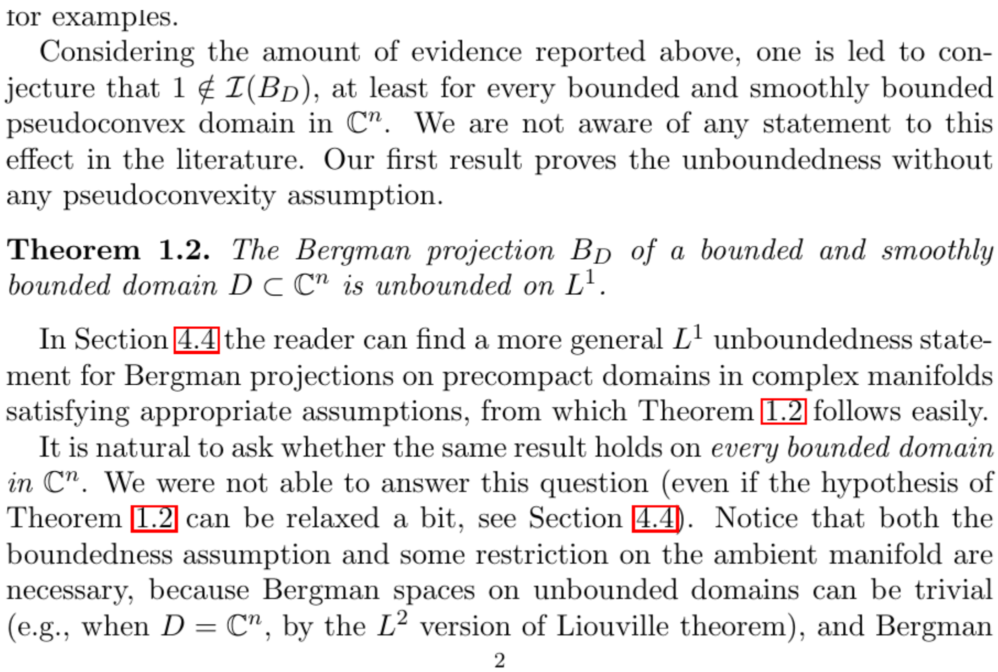
Automatic math-discovery packet for arXiv:2104.04009

August 9, 2026

Classification: candidate complete theorem for the simply connected planar case; partial toward the source question. Every simply connected planar domain of finite area has an L^1-unbounded Bergman projection. No boundary regularity, Jordan hypothesis, or boundary accessibility assumption is used.

Source question

The source is G. M. Dall'Ara, *Around L^1 (un)boundedness of Bergman and Szegő projections*, arXiv:2104.04009, published in the *Journal of Functional Analysis* 283 (2022), Paper No. 109550. It proves that the Bergman projection of every bounded smoothly bounded domain in $\mathbb{C}^n$ is unbounded on L^1, and then asks whether the same is true for every bounded domain in $\mathbb{C}^n$.



for examples.

Considering the amount of evidence reported above, one is led to conjecture that $1 \notin \mathcal{I}(B_D)$, at least for every bounded and smoothly bounded pseudoconvex domain in $\mathbb{C}^n$. We are not aware of any statement to this effect in the literature. Our first result proves the unboundedness without any pseudoconvexity assumption.

Theorem 1.2. *The Bergman projection B_D of a bounded and smoothly bounded domain $D \subset \mathbb{C}^n$ is unbounded on L^1.*

In Section 4.4 the reader can find a more general L^1 unboundedness statement for Bergman projections on precompact domains in complex manifolds satisfying appropriate assumptions, from which Theorem 1.2 follows easily.

It is natural to ask whether the same result holds on *every bounded domain in $\mathbb{C}^n$*. We were not able to answer this question (even if the hypothesis of Theorem 1.2 can be relaxed a bit, see Section 4.4). Notice that both the boundedness assumption and some restriction on the ambient manifold are necessary, because Bergman spaces on unbounded domains can be trivial (e.g., when $D = \mathbb{C}^n$, by the L^2 version of Liouville theorem), and Bergman

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Figure 1: The theorem and open question on page 2 of the source PDF.

Here area measure is unnormalized Lebesgue measure dA, and $\mathcal{B}_\Omega$ denotes the $L^2(\Omega)$ Bergman projection. Boundedness on $L^1(\Omega)$ means that the restriction to $L^1(\Omega) \cap L^2(\Omega)$ admits a bounded extension to $L^1(\Omega)$.

Main result

Theorem 1 (finite-area simply connected domains)**.** *Let $\Omega \subset \mathbb{C}$ be a simply connected domain with $0 < A(\Omega) < \infty$. Then $\mathcal{B}_\Omega$ is unbounded on $L^1(\Omega)$.*

Corollary 1. *Every bounded simply connected domain in $\mathbb{C}$, with completely arbitrary boundary, has an L^1-unbounded Bergman projection.*

The finite-area formulation is slightly stronger than the bounded-domain subcase of the source question. It includes non-Jordan domains, domains with fractal boundary, and domains with no rectifiable boundary arc.

Proof intuition

Let $\psi : \mathbb{D} \to \Omega$ be a Riemann map. Conformal change of variables turns $\mathcal{B}_\Omega$ into the ordinary disk Bergman projection acting on $L^1(\mathbb{D}, w\, dA)$, where $w = |\psi'|$. Endpoint boundedness would force every weighted kernel column to satisfy

$$\int_{\mathbb{D}} \frac{w(z)}{\pi|1 - z\overline{a}|^2}\, dA(z) \le Cw(a). \tag{1}$$

There is a two-step feedback contradiction. First, the disk kernel is bounded below everywhere, so (1) forces w to have a positive global lower bound. Feeding that lower bound back into (1) and integrating the unweighted kernel exactly forces

$$w(a) \gtrsim \frac{-\log(1 - |a|^2)}{|a|^2},$$

uniformly in the argument of a. Thus $|\psi'|$ tends uniformly to infinity on concentric circles. But ψ' never vanishes, so $1/\psi'$ is analytic; the maximum principle would then force $1/\psi'$ to vanish identically.

Conformal transfer

The disk Bergman kernel is

$$K_{\mathbb{D}}(z, u) = \frac{1}{\pi(1 - z\overline{u})^2}.$$

Put $w = |\psi'|$ and define

$$Uf = (f \circ \psi)\psi'.$$

The change-of-variables formula gives

$$\|Uf\|_{L^2(\mathbb{D})} = \|f\|_{L^2(\Omega)}, \qquad \|Uf\|_{L^1(\mathbb{D}, w\, dA)} = \|f\|_{L^1(\Omega)}. \tag{2}$$

The transformation rule for Bergman kernels gives, on L^2,

$$U\mathcal{B}_\Omega U^{-1} = \mathcal{B}_{\mathbb{D}}. \tag{3}$$

Hence L^1-boundedness of $\mathcal{B}_\Omega$ would imply boundedness of $\mathcal{B}_{\mathbb{D}}$ on $L^1(\mathbb{D}, w\, dA)$.

Since ψ is injective, the area formula and Cauchy–Schwarz give

$$\int_{\mathbb{D}} |\psi'(z)|^2\, dA(z) = A(\Omega) < \infty, \qquad 0 < \int_{\mathbb{D}} w\, dA < \infty. \tag{4}$$

The weighted endpoint obstruction

Lemma 1 (necessary column estimate)**.** *Let w be positive and continuous on $\mathbb{D}$. If $\mathcal{B}_{\mathbb{D}}$ is bounded on $L^1(\mathbb{D}, w\,dA)$ with norm at most C, then for every $a \in \mathbb{D}$,*

$$I_w(a) := \int_{\mathbb{D}} |K_{\mathbb{D}}(z,a)| w(z)\,dA(z) \le Cw(a). \tag{5}$$

Proof. For sufficiently small $\varepsilon > 0$, let

$$h_\varepsilon(u) = \frac{\mathbf{1}_{B(a,\varepsilon)}(u)}{A(B(a,\varepsilon))}.$$

Then $h_\varepsilon \in L^2(\mathbb{D}) \cap L^1(\mathbb{D}, w\,dA)$,

$$\|h_\varepsilon\|_{L^1(w)} = \frac{1}{A(B(a,\varepsilon))} \int_{B(a,\varepsilon)} w(u)\,dA(u) \longrightarrow w(a),$$

and $\mathcal{B}_{\mathbb{D}} h_\varepsilon(z) \to K_{\mathbb{D}}(z,a)$ pointwise. The operator bound and Fatou's lemma yield (5). □

Theorem 2 (zero-free analytic weights never work)**.** *Let g be a zero-free analytic function on $\mathbb{D}$ such that*

$$0 < \int_{\mathbb{D}} |g|\,dA < \infty.$$

Then $\mathcal{B}_{\mathbb{D}}$ is unbounded on $L^1(\mathbb{D}, |g|\,dA)$.

Proof. Suppose the contrary, write $w = |g|$, and let C be an operator bound. Set

$$M = \int_{\mathbb{D}} w\,dA > 0.$$

Because $|1 - z\overline{a}| \le 2$ for $z, a \in \mathbb{D}$, we have

$$I_w(a) \ge \frac{M}{4\pi}.$$

Combining this with (5) gives the uniform positive lower bound

$$w(a) \ge m := \frac{M}{4\pi C} > 0 \qquad (a \in \mathbb{D}). \tag{6}$$

Use (6) once more inside the column integral. For $a \ne 0$, orthogonality of the monomials (or direct angular integration) gives

$$\begin{aligned} \frac{1}{\pi} \int_{\mathbb{D}} \frac{dA(z)}{|1 - z\overline{a}|^2} &= \sum_{n=0}^{\infty} \frac{|a|^{2n}}{n+1} \\ &= \frac{-\log(1 - |a|^2)}{|a|^2}. \end{aligned} \tag{7}$$

Therefore (5)–(7) imply

$$w(a) \ge \frac{m}{C}, \frac{-\log(1 - |a|^2)}{|a|^2}. \tag{8}$$

The right side tends to infinity with $|a|$, independently of the argument of a.

Since g has no zeros, $1/g$ is analytic on $\mathbb{D}$. From (8), for $0 < r < 1$,

$$\max_{|z|=r} \left| \frac{1}{g(z)} \right| \le \frac{C}{m}, \frac{r^2}{-\log(1 - r^2)} \longrightarrow 0.$$

The maximum principle now gives

$$\left| \frac{1}{g(0)} \right| \le \max_{|z|=r} \left| \frac{1}{g(z)} \right| \longrightarrow 0,$$

which is impossible. This proves the theorem. □

Proof of the main theorem

Proof of Theorem 1. Let $\psi : \mathbb{D} \to \Omega$ be a Riemann map. Its derivative is zero-free, and (4) shows that $|\psi'|$ is integrable. If $\mathcal{B}_\Omega$ were bounded on $L^1(\Omega)$, (2)–(3) would make $\mathcal{B}_\mathbb{D}$ bounded on $L^1(\mathbb{D}, |\psi'|dA)$, contradicting Theorem 2. $\square$

Hostile verification audit

Potential failure	Resolution
Wrong conformal weight	The multiplier ψ' in U cancels one of the two Jacobian factors, leaving exactly $|\psi'|$ in the L^1 norm.
Weight not integrable	Finite area gives $\psi' \in A^2(\mathbb{D})$, hence $\psi' \in A^1(\mathbb{D})$ by Cauchy–Schwarz.
Illegitimate point mass	Lemma 1 uses normalized indicators in $L^1(w) \cap L^2$, kernel continuity, and Fatou's lemma.
Lower-bound direction reversed	The global estimate $|K_\mathbb{D}(z,a)| \geq 1/(4\pi)$ combines with $I_w(a) \leq Cw(a)$ to give $w(a) \geq M/(4\pi C)$.
Incorrect kernel integral	Expanding both geometric series leaves only equal powers after angular integration and yields exactly the series in (7).
Growth only almost everywhere	The column estimate holds at every interior point because w is continuous; (8) is uniform on each circle.
Reciprocal may be singular	A conformal map has nonvanishing derivative, so $1/\psi'$ is analytic throughout $\mathbb{D}$.

Scope, search status, and novelty caution

This completely removes boundary assumptions in the simply connected planar finite-area case, but it does not answer the source question for multiply connected planar domains or domains in $\mathbb{C}^n$ with $n \geq 2$. The earlier rectifiable-boundary-arc version of this packet is strictly subsumed by Theorem 1.

Bounded searches through August 9, 2026 included the exact source question, the source article and its citations, exact phrases combining simply connected planar domains with strong L^1 Bergman boundedness, and recent weighted and weak-type work. No prior statement of Theorem 1 or Theorem 2 was located. This is a bounded novelty check, not a claim of priority. The recent paper [2] treats weak-type endpoint estimates on simply connected planar domains but does not state the strong L^1 theorem proved here.

Human-review recommendation

Review as a likely valid, high-value partial result. The proof is short and self-contained after standard conformal covariance. The highest-value checks are the single-factor weight in (2), the L^1 kernel-column test, and the maximum-principle contradiction following the exact integral (7).

References

[1] G. M. Dall'Ara, *Around L^1 (un)boundedness of Bergman and Szegő projections*, J. Funct. Anal. 283 (2022), Paper No. 109550; arXiv:2104.04009.

[2] A. W. Green, C. B. Stockdale, B. Sweeting, and N. A. Wagner, *Weak-type estimates for the Bergman projection on planar domains*, arXiv:2607.09642v1 (2026).